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Branch: B.Tech CSE

Subject: Linux System Programming Lab

OSSP_Practical1&2

OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical1.c *
#include <stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
char cmd[100];
printf("Enter Linux command: ");
scanf("%s", cmd);
pid_t pid = fork();
if(pid == 0)
{
printf("Child PID : %d\n", getpid());
execlp(cmd, cmd, NULL);
perror("Execution Failed");
}
else if(pid > 0)
{
printf("Parent PID : %d\n", getpid());
wait(NULL);
printf("Child process completed.\n");
}
else
{
printf("Fork failed\n");
}
return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_/ Go To Line M-E Redo
```

```
Write to File: practical1.c
^G Help      M-D DOS Format M-A Append   M-B Backup File ^T Browse
^C Cancel    M-M Mac Format M-P Prepend  ^Q Discard buffer
```

```
Select monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~$ mkdir OSSP-Practical
monis@DESKTOP-C3GM4T6:~$ cd OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical1.c
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical1.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical1.c -o practical1
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical1
Enter Linux command: date
Parent PID : 997
Child PID : 1016
Fri Aug 7 09:46:54 UTC 2026
Child process completed.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical1.c -o practical1
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical1
Enter Linux command: pwd
Parent PID : 1046
Child PID : 1047
/home/monis/OSSP-Practical
Child process completed.
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical1.c -o practical1
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical1
Enter Linux command: ls
Parent PID : 1094
Child PID : 1095
practical1 practical1.c
Child process completed.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ uname
Linux
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ uname -a
Linux DESKTOP-C3GM4T6 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 GNU/Linux
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            39 bits physical, 48 bits virtual
Byte Order:              Little Endian
CPU(s):                   8
On-line CPU(s) list:     0-7
Vendor ID:               GenuineIntel
Model name:              Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
CPU family:              6
Model:                   142
Thread(s) per core:      2
Core(s) per socket:      4
Socket(s):               1
Stepping:                12
BogoMIPS:                4224.01
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
                        h mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon
                        rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16
                        pdcm pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor_l
                        ahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                        d ept_ad fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt
                        xsaveopt xsavec xgetbv1 xsaves vnmmls md_clear flush_l1d arch_capabilities

Virtualization features:
Virtualization:          VT-x
Hypervisor vendor:       Microsoft
Virtualization type:     full
Caches (sum of all):
L1d:                     128 KiB (4 instances)
L1i:                     128 KiB (4 instances)
L2:                      1 MiB (4 instances)
L3:                      6 MiB (1 instance)
NUMA:
NUMA node(s):            1
NUMA node0 CPU(s):      0-7
Vulnerabilities:
Gather data sampling:    Unknown: Dependent on hypervisor status
Itlb multihit:           KVM: Mitigation: VMX disabled
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:         Vulnerable: Clear CPU buffers attempted, no microcode; SMT Host state unknown
Reg file data sampling:  Not affected
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBRSB-eI
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
Tsx async abort:      Not affected
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.6M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0    1G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ps
PID TTY          TIME CMD
top - 09:54:05 up 32 min,  1 user,  load average: 0.04, 0.01, 0.00
Tasks: 27 total,  1 running, 26 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3790.2 total, 3249.1 free,  499.3 used,  177.4 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,   0.0 used. 3291.0 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24316  14284  10956 S   0.0   0.4   0:02.23 systemd
    2 root        20   0    3120   2048   1920 S   0.0   0.1   0:00.07 init-systemd(Ub
    6 root        20   0    3120   1792   1792 S   0.0   0.0   0:00.00 init
   49 root       19  -1   50380  15872  14848 S   0.0   0.4   0:00.54 systemd-journal
   83 systemd+    20   0   22524  14080  11520 S   0.0   0.4   0:00.15 systemd-resolve
   92 root        20   0   35104  11648   8832 S   0.0   0.3   0:01.68 systemd-udev
  112 root        20   0    2880   1792   1792 S   0.0   0.0   0:00.05 chronyd-starter
  113 root        20   0    4464   2816   2688 S   0.0   0.1   0:00.02 cron
  128 message+    20   0    8800   4864   4352 S   0.0   0.1   0:00.12 dbus-daemon
  152 root        20   0   44076  28604  13824 S   0.0   0.7   0:00.40 networkd-dispat
  160 root        20   0   18420   8704   7680 S   0.0   0.2   0:00.15 systemd-logind
  173 syslog      20   0  220532   5120   4224 S   0.0   0.1   0:00.14 rsyslogd
  210 _chrony      20   0   24248  11264   6784 S   0.0   0.3   0:00.11 chronyd
  219 _chrony      20   0   12076   2240   1664 S   0.0   0.1   0:00.01 chronyd
  238 root        20   0    5528   2688   2560 S   0.0   0.1   0:00.02 agetty
  241 root        20   0  123440  31832  17408 S   0.0   0.8   0:00.36 unattended-upgr
  262 root        20   0    5484   2560   2432 S   0.0   0.1   0:00.02 agetty
  307 root        20   0    3136    896    768 S   0.0   0.0   0:00.00 SessionLeader
  308 root        20   0    3136   1288   1152 S   0.0   0.0   0:00.27 Relay(310)
  310 monis       20   0    6260   5376   3712 S   0.0   0.1   0:00.18 bash
  314 root        20   0    8632   5376   4608 S   0.0   0.1   0:00.01 login
  373 monis       20   0   22320  13184  10624 S   0.0   0.3   0:00.14 systemd
  377 monis       20   0   22892   4044   2048 S   0.0   0.1   0:00.00 (sd-pam)
  405 monis       20   0    6128   4992   3712 S   0.0   0.1   0:00.03 bash
  818 polkitd     20   0  306436   8064   7424 S   0.0   0.2   0:00.10 polkitd
 1172 monis       20   0    8336   5888   3712 R   0.0   0.2   0:00.12 top
 1183 root        20   0 1866280  14592  11904 S   0.0   0.4   0:00.08 wsl-pro-service
```

Press q

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
262 root      20    0    5484    2560    2432 S    0.0    0.1    0:00.02 agetty
307 root      20    0    3136     896     768 S    0.0    0.0    0:00.00 SessionLeader
310 monis     20    0    6260    5376    3712 S    0.0    0.1    0:00.18 bash
314 root      20    0    8632    5376    4608 S    0.0    0.1    0:00.01 login
373 monis     20    0    22320   13184   10624 S    0.0    0.3    0:00.14 systemd
377 monis     20    0    22892    4044    2048 S    0.0    0.1    0:00.00 (sd-pam)
405 monis     20    0     6128    4992    3712 S    0.0    0.1    0:00.03 bash
818 polkitd   20    0   306436    8064    7424 S    0.0    0.2    0:00.11 polkitd
1172 monis     20    0     8336    5888    3712 R    0.0    0.2    0:00.18 top
1183 root      20    0   1866280  14592   11904 S    0.0    0.4    0:00.09 wsl-pro-service

monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

-
2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical2.c *
#include<stdio.h>
#include<fcntl.h>
#include<unistd.h>

int main()
{
int src,dest;
char buffer[1024];
int bytes;
src = open("input.txt", O_RDONLY);

if(src < 0)
{
printf("Cannot open source file\n");
return 1;
}
dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

while((bytes = read(src, buffer, sizeof(buffer))) > 0)
{
write(dest, buffer, bytes);
}
close(src);
close(dest);

printf("file copied successfully.\n");
return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^J Justify    ^/ Go To Line M-E Redo
```

```
Write to File: practical2.c
^G Help      M-D DOS Format M-A Append    M-B Backup File ^T Browse
^C Cancel    M-M Mac Format M-P Prepend   ^Q Discard buffer
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical2.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical2.c -o practical2
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical$ nano practical2.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical2.c -o practical2
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano input.txt
```

```
GNU nano 8.7.1 input.txt *
Hello Everone
Welcome to OSSP Lab
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
Write to File: input.txt
```

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^O Discard buffer	

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical2.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical2.c -o practical2
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical2
file copied successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat output.txt
Hello Everone
Welcome to OSSP Lab
```

Run strace

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical2.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc practical2.c -o practical2
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical2
file copied successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat output.txt
Hello Everone
Welcome to OSSP Lab
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ strace cat input.txt
Command 'strace' not found, but can be installed with:
sudo apt install strace
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 2: If strace is not installed

You may get:

strace: command not found

Install it:

sudo apt update

sudo apt install strace

Then run:

strace cat input.txt

Just install it once.

Step 1: Update package list

sudo apt update

Enter your password if prompted.

Step 2: Install strace

sudo apt install strace

When asked:

Do you want to continue? [Y/n]

Type:

Y

and press **Enter**.

Step 3: Verify the installation

`strace --version`

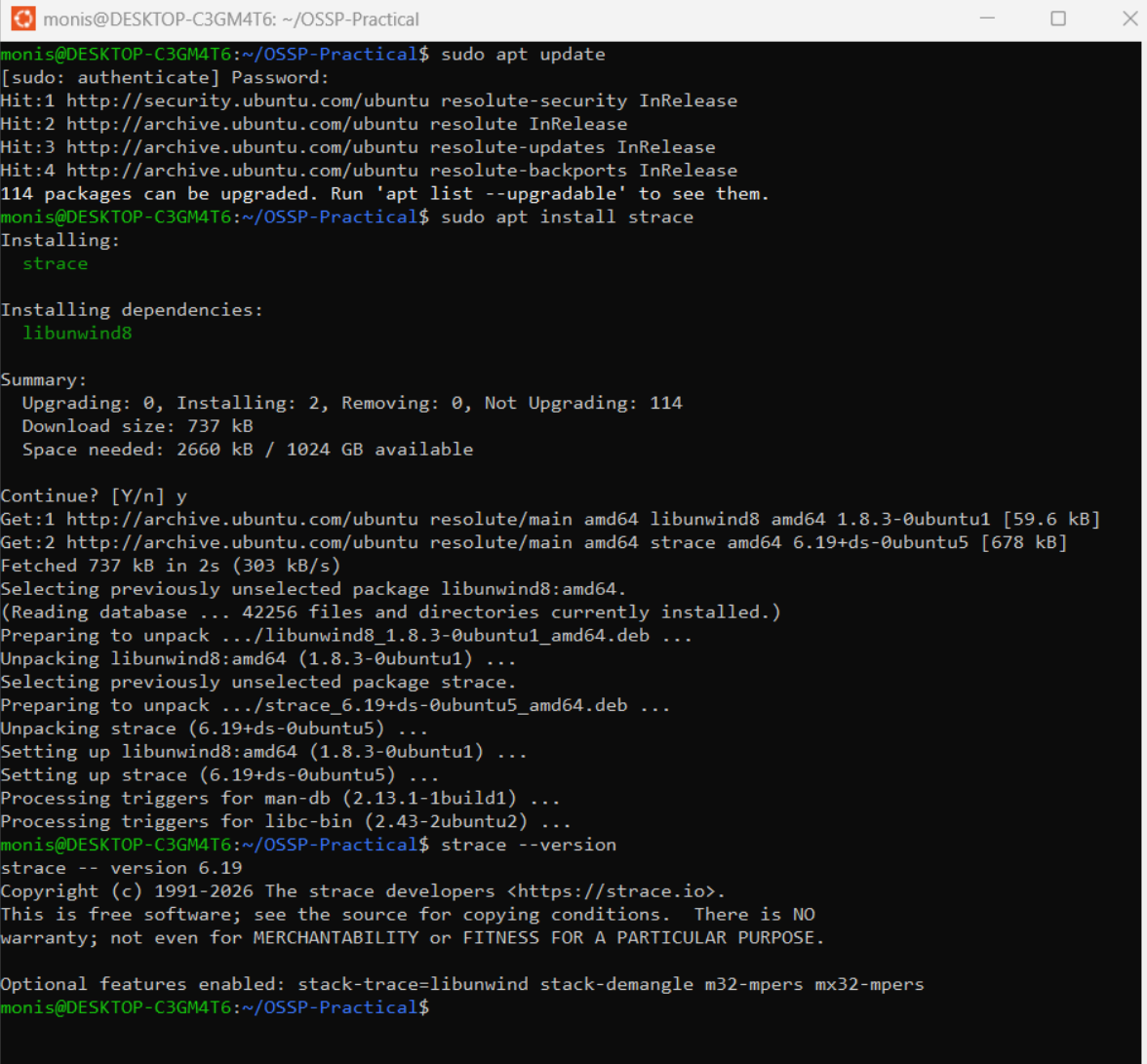
You should see something like:

`strace -- version 6.x`

Step 4: Run the practical command

Since your file is **input.txt**, run:

`strace cat input.txt`



```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
114 packages can be upgraded. Run 'apt list --upgradable' to see them.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ sudo apt install strace
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 114
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 2s (303 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 42256 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mpers
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

monis@DESKTOP-C3GM4T6: ~/OSSP-Practical

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mpers

monis@DESKTOP-C3GM4T6:~/OSSP-Practical\$ strace cat input.txt

execve("/usr/bin/cat", ["cat", "input.txt"], 0x7ffd2125df88 /* 26 vars */) = 0

brk(NULL) = 0x586b73c9c000

mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x75241484e000

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

fstat(3, {st_mode=S_IFREG|0644, st_size=20025, ...}) = 0

mmap(NULL, 20025, PROT_READ, MAP_PRIVATE, 3, 0) = 0x752414849000

close(3) = 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0

mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x752414816000

mmap(0x75241481d000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x75241481d000

mmap(0x75241483e000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x75241483e000

mmap(0x752414845000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x752414845000

mmap(0x752414847000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x752414847000

close(3) = 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0

mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7524147e8000

mmap(0x7524147ec000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7524147ec000

mmap(0x752414810000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x752414810000

mmap(0x752414814000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x752414814000

close(3) = 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0

mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7524146c2000

mmap(0x7524146d4000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x7524146d4000

mmap(0x752414774000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x752414774000

mmap(0x7524147e6000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x7524147e6000

close(3) = 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

pread64(3, "\6\0\0\0\4\0"..., 840, 64) = 840

fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0

pread64(3, "\6\0\0\0\4\0"..., 840, 64) = 840

mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x752414400000

mmap(0x752414428000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x752414428000

mmap(0x7524145c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x7524145c0000

mmap(0x75241460e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x75241460e000

mmap(0x752414614000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x752414614000

close(3) = 0

openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0

mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x752414354000

mmap(0x752414357000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x752414357000

mmap(0x7524143d3000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x7524143d3000